# Privacy-Preserving Fuzzy Matching in Practical Contexts

how to gracefully panic when real-world data is far from exact

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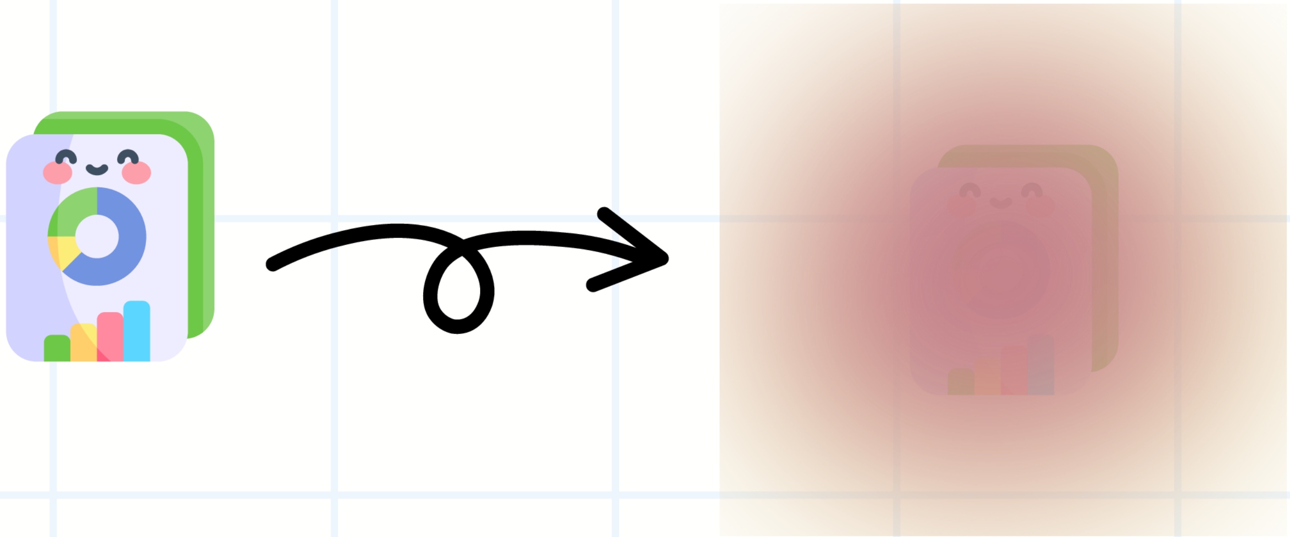
# Motivation

Why Privacy-Preserving Fuzzy Matching?



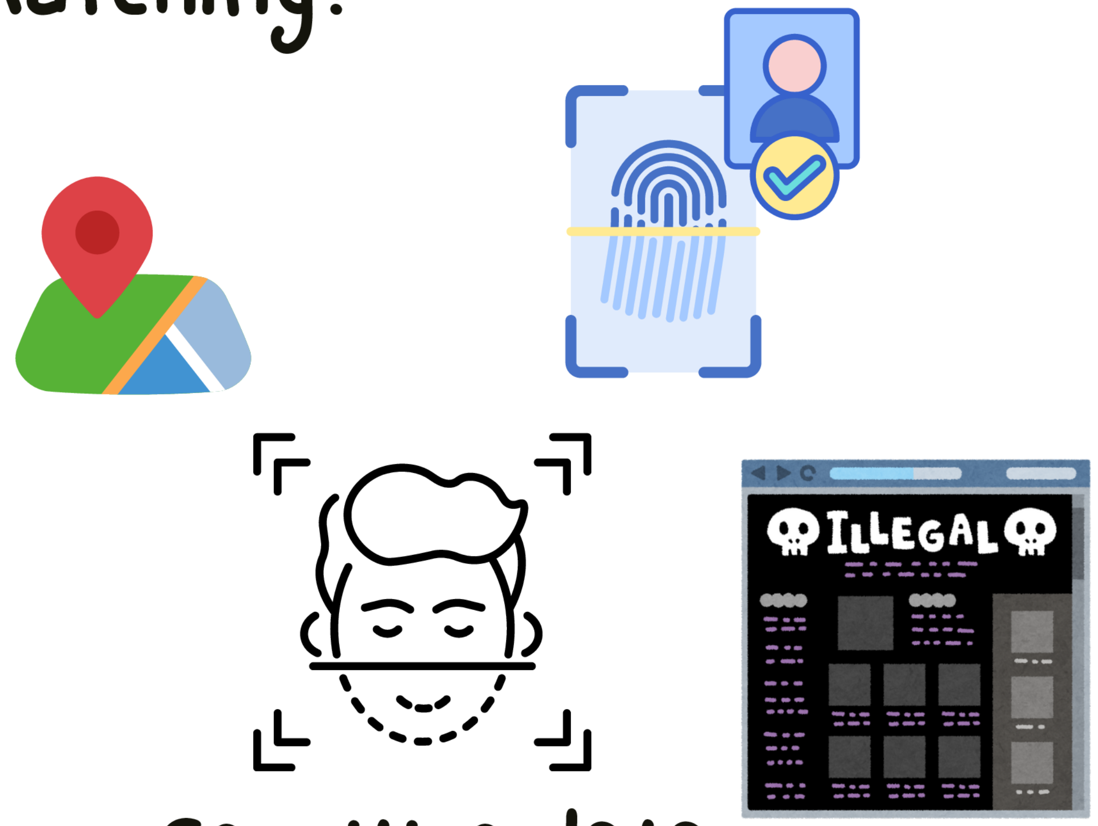
Noisy data

Information can have errors or inconsistencies, variations, etc.

# Motivation

## Why Privacy-Preserving Fuzzy Matching?

Sensitive datasets like biometrics, location history must remain confidential.




Sensitive data

# Motivation

Classic exact matching is not enough...

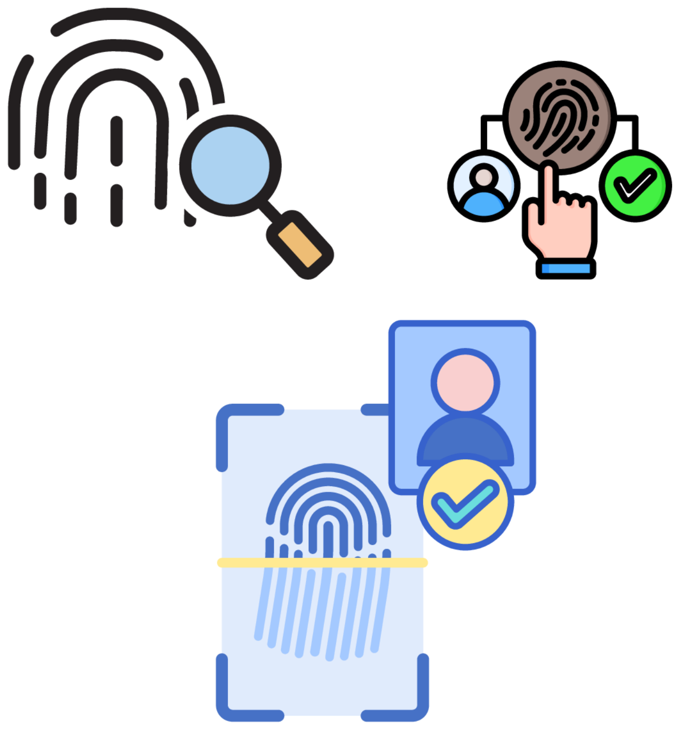
Most existing protocols consider data as exact items, so we need new protocols...

Two slightly different informations might still correspond to the same one

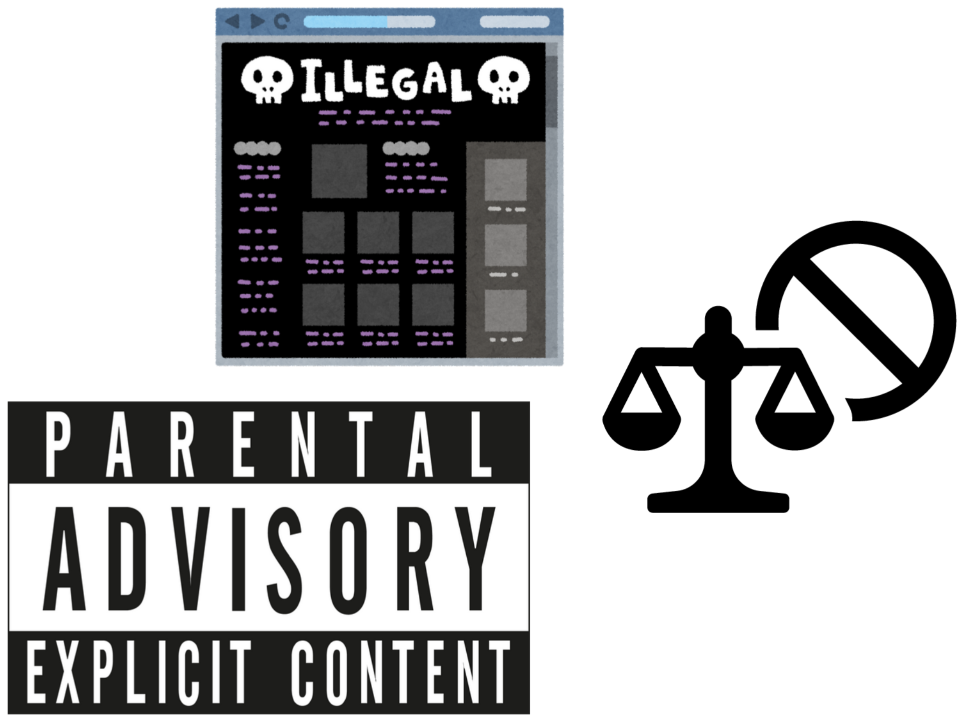
What if we try matching elements when they are "close" with respect to some distance metric...?

# Real World Applications

Contact Tracing



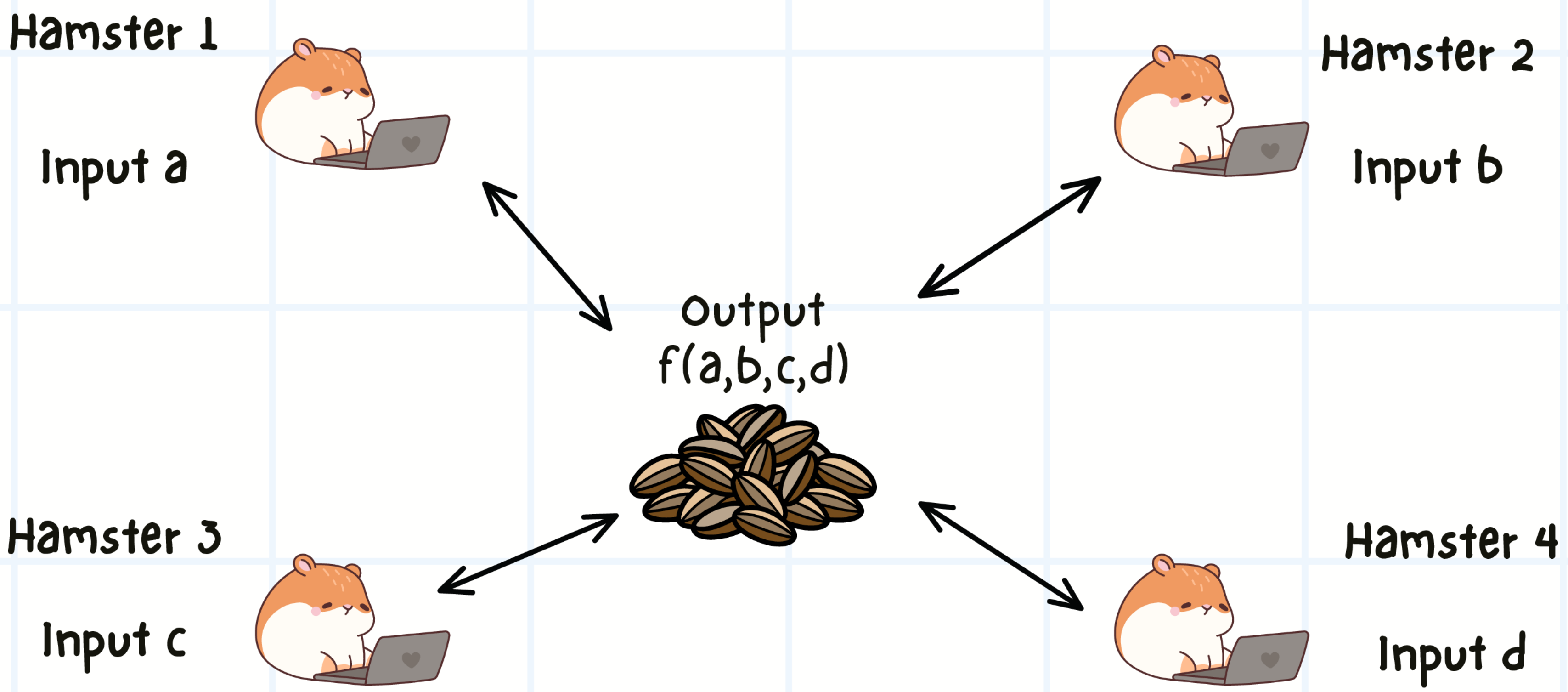
Biometric Search




Detection of sensitive media

# Background

What is Secure Multi-Party Computation?



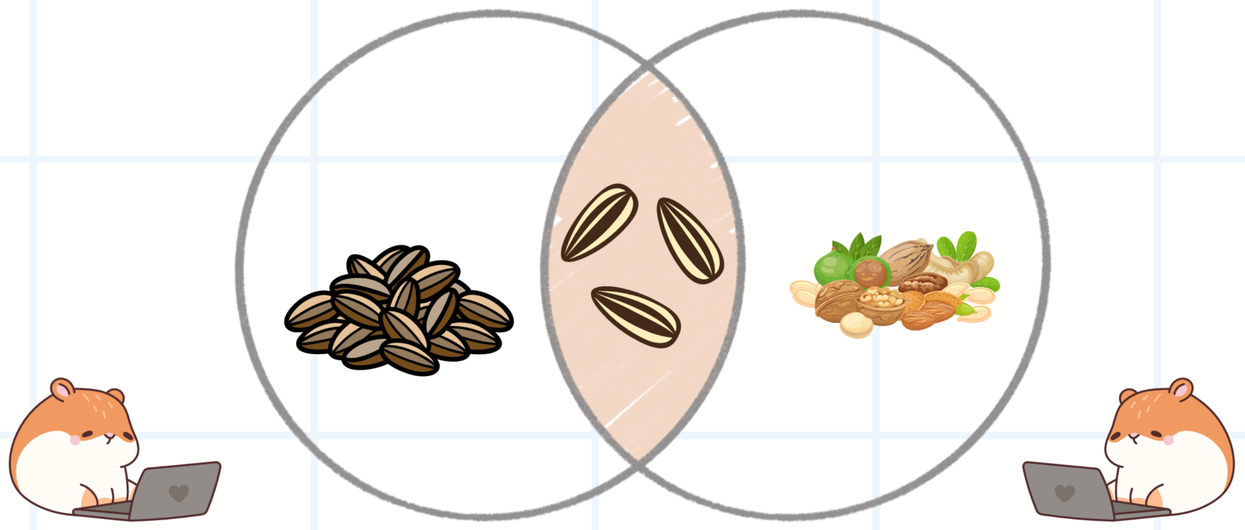

Multiple parties, each with a private inputs, collaborate to compute a function over their combined data

Nothing else is revealed except the output.

Everyone follows the protocol and are ideally "honest"

# Background

What is Private Set Intersection?



Two parties wish to compute $X \cap Y$
without revealing their individual set

Parties may only learn the common elements
and nothing else

# Background

## What is Fuzzy Matching?

Hamster ≈ Hasmter

Allows small differences between data elements

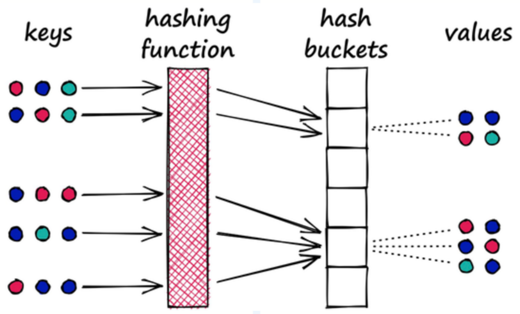
Matches are based on a similarity score or distance measure, not strict identity.

Find items that are "close enough" according to a threshold.

# Goal

How to Build Privacy-Preserving Fuzzy Matching?

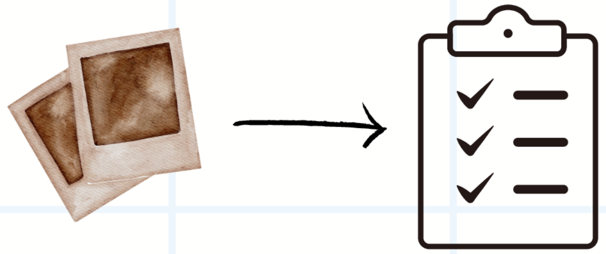
raw data → Approximate encoding (Locality sensitive hash or threshold projections) → Secure Comparision (homomorphic encryption or secret share) → Similarities search within some threshold (Private Approximate Membership Comparison)




# Apple PSI system [BBM+21]

## Privacy-Preserving Detection for CSAM (Child Sexual Abuse Material)

Apple uses a threshold Private Set Intersection with Associated Data protocol



User only uploads encrypted safety vouchers – no raw images

API runs locally on-device using preloaded encrypted data

Cloud never sees user photos unless there's a confirmed threshold breach

[BBM+21]. Abhishek Bhowmick, Dan Boneh, Steve Myers, Kunal Talwar, and Karl Tarbe. The apple psi system, 2021.

# Apple PSI system [BBM+21]

## Privacy-Preserving Detection for CSAM (Child Sexual Abuse Material)

User's Iphone

↓ Safety voucher

Apple Cloud

↓

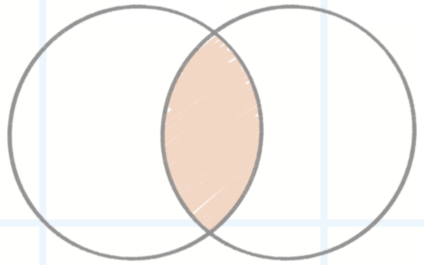
only decrypts if threshold is met

IMPORTANT Detection happens privately, unless evidence crosses a threshold.

No decryption unless there's convincing evidence – even then, no full dataset is revealed. IMPORTANT

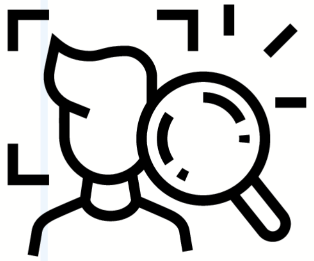
[BBM+21]. Abhishek Bhowmick, Dan Boneh, Steve Myers, Kunal Talwar, and Karl Tarbe. The apple psi system, 2021.

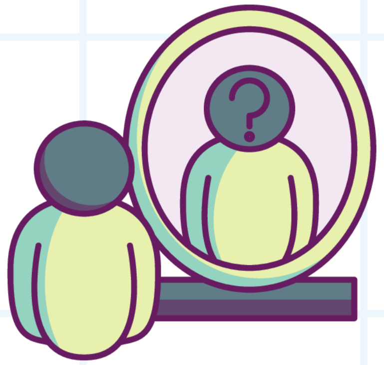
# Fuzzy Labeled PSI for Biometrics [UCK+21]



Combines fuzzy matching with private set intersection

Designed for real-time biometric search (e.g., faces in surveillance footage)





The client learns the label (e.g., identity) of matching database items – without revealing their input

Server learns nothing about the query

[UCK+21]. Erkam Uzun, Simon P. Chung, Vladimir Kolesnikov, Alexandra Boldyreva, and Wenke Lee. Fuzzy labeled private set intersection with applications to private real-time biometric search. In Michael Bailey and Rachel Greenstadt, editors, USENIX Security 2021: 30th USENIX Security Symposium, pages 911–928. USENIX Association, August 11–13, 2021.

# Fuzzy Labeled PSI for Biometrics [UCK+21]

client's face

↓

Encrypted query

↓

Sent to Cloud

↓ Threshold Matching to return encrypted label

Label displayed to client

Hamster 1

transforms fuzzy biometric matching into exact share-based comparisons, using binary encoding, private subsampling, and threshold secret sharing.

[UCK+21]. Erkam Uzun, Simon P. Chung, Vladimir Kolesnikov, Alexandra Boldyreva, and Wenke Lee. Fuzzy labeled private set intersection with applications to private real-time biometric search. In Michael Bailey and Rachel Greenstadt, editors, USENIX Security 2021: 30th USENIX Security Symposium, pages 911–928. USENIX Association, August 11–13, 2021.

# Private Approximate Media Matching [KM21]

Detecting Harmful Media- Privately

E2EE limits traditional CSAM and harmful content detection.

We wish to check if a media file approximately matches a harmful item in a private set

Private Approximate Membership Computation (PAMC) to allow encrypted matching of near-identical media, without exposing the content or the reference set.

[KM21].Anunay Kulshrestha and Jonathan Mayer. Identifying harmful media in {Endto-End} encrypted communication: Efficient private membership computation. In 30th USENIX Security Symposium (USENIX Security 21), pages 893–910, 2021.

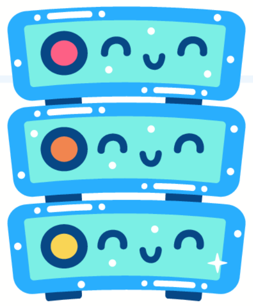
# Private Approximate Media Matching [KM21]

## Detecting Harmful Media- Privately

**user media**

PHF (Percetual Hash Function)
+
FHE

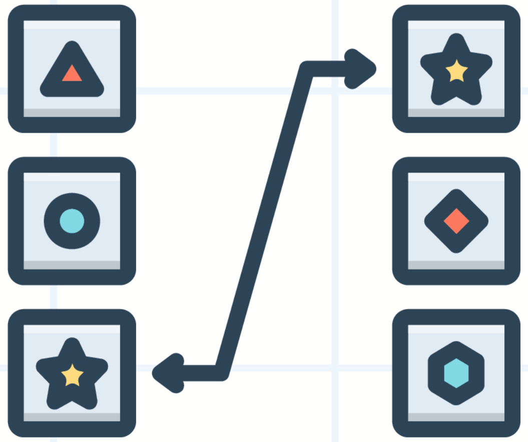
try to imitate the human perception of media by giving the same hash value for those with some semantic similarity



**server**

runs a protocol of private approximate membership over the encrypted media

calculate distance between data (hamming)



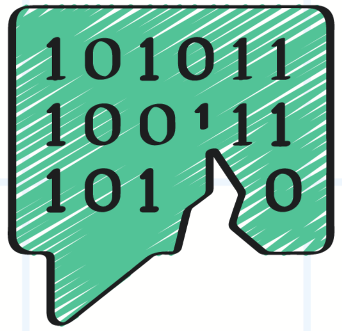
Match if distance between data is inside some threshold

[KM21].Anunay Kulshrestha and Jonathan Mayer. Identifying harmful media in {Endto-End} encrypted communication: Efficient private membership computation. In 30th USENIX Security Symposium (USENIX Security 21), pages 893–910, 2021.

# Challenges, Open Questions and more

## Security & Leakage Risks

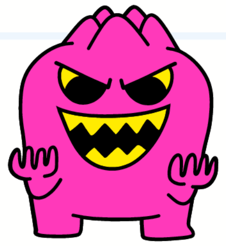
Protocols may leak access patterns, approximate distances, or set membership

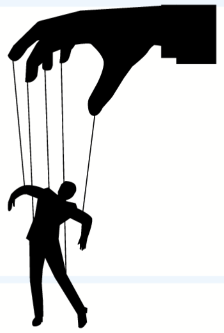


Attack examples:
[HWAM24] Evading PHFs by generating near-looking but non-matching inputs
[Teague+24] Attacks on client-side scanning and PSI integrity checks

## Protocol Model Gaps

Many systems assume honest-but-curious servers – what about malicious or coerced service providers?

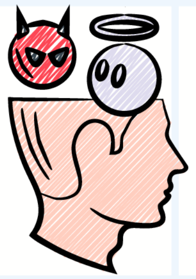




Traceability proposals like [Bartusek+23] show how easily servers could overstep its role and perform some "power abuse"

## Can we get more efficient, please?

Fuzzy PSI remains cryptographically heavy



Efficiency and leakage are hard to make better together – but so is privacy and moderation


[Bartusek+23] Bartusek, J., Garg, S., Jain, A., Policharla, GV. (2023). End-to-End Secure Messaging with Traceability Only for Illegal Content. In: Hazay, C., Stam, M. (eds) Advances in Cryptology – EUROCRYPT 2023. EUROCRYPT 2023. Lecture Notes in Computer Science, vol 14008. Springer, Cham. https://doi.org/10.1007/978-3-031-30589-4_2
[Teague+24].S. Vanessa Teague and Shaanan Cohney, "Techniques for detecting known illegal material" in IETF Brisbane March 20, 2024
[HWAM24].S. Hawkes, C. Weinert, T. Almeida and M. Mehrnezhad, "Perceptual Hash Inversion Attacks on Image-Based Sexual Abuse Removal Tools," in IEEE Security & Privacy, doi: 10.1109/MSEC.2024.3485497.

# Challenges, Open Questions and more

## Backdoors & Traceability Debate

Can we build just enough visibility into E2EE systems?

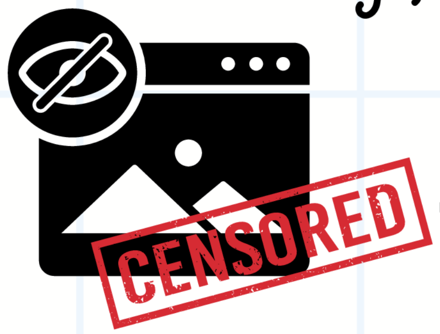
In [Bartusek+23], most protocols rely on trusted setup and are hard to verify publicly


## Whom do protocols serve?

Detection systems are often built without consent or transparency...

What if matching is biased or abused?

→ False flags, surveillance, censorship




From [Kulshrestha+21]: PAMC flags "similar" media – but adversarial noise can fool PHFs or trigger false matches

[KM21].Anunay Kulshrestha and Jonathan Mayer. Identifying harmful media in {Endto-End} encrypted communication: Efficient private membership computation. In 30th USENIX Security Symposium (USENIX Security 21), pages 893–910, 2021.

[Bartusek+23] Bartusek, J., Garg, S., Jain, A., Policharla, GV. (2023). End-to-End Secure Messaging with Traceability Only for Illegal Content. In: Hazay, C., Stam, M. (eds) Advances in Cryptology – EUROCRYPT 2023. EUROCRYPT 2023. Lecture Notes in Computer Science, vol 14008. Springer, Cham. https://doi.org/10.1007/978-3-031-30589-4_2

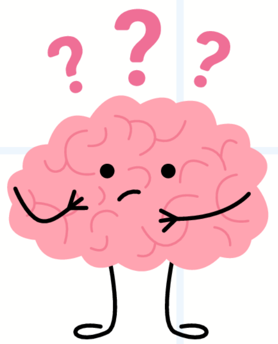
# Challenges, Open Questions and more

## Are we any close to Standardization?

PSI standards is emerging (WPEC, NIST) –
but fuzzy PSI is still immature

[Steve Lu @ WPEC 2024] Is Approximate PSI ready for standardization?


How should systems prove fairness and verifiability?




Can PSI-based matching be used in public interest tech (e.g., whistleblowing, journalism)?

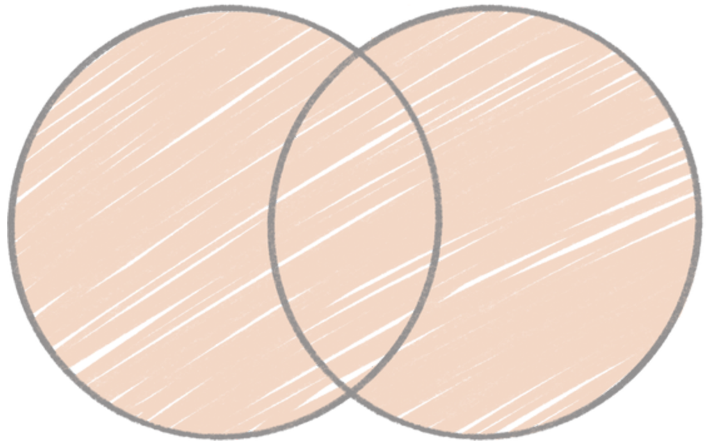
Can we audit PHF-based matching?

How to allow user consent for detection?


[Steve Lu @ WPEC 2024].Steve Lu, CEO at Stealth Software technologies, Inc. "Paths Toward PSI Standardization and Approximate PSI "In NIST Workshop on Privacy-Enhancing Cryptography 2024 2024-09-24 (PSI Day)

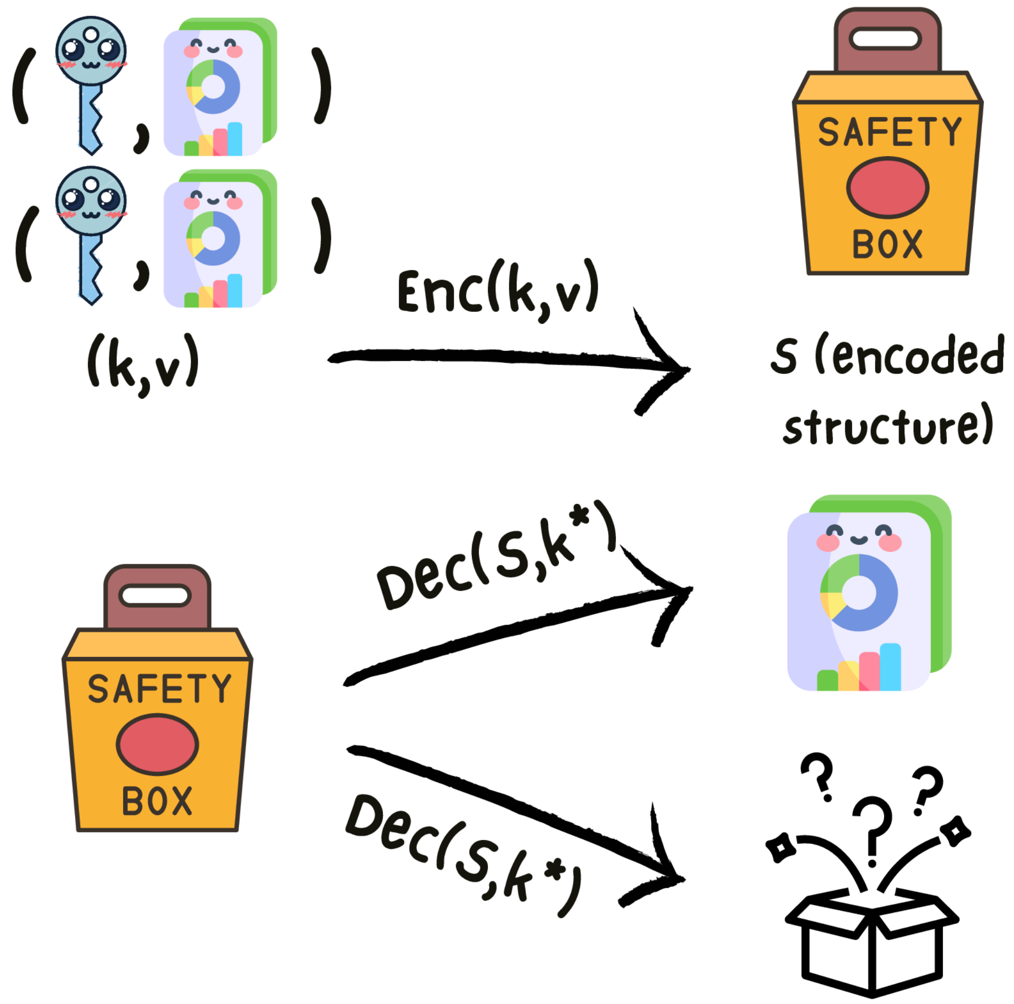
# Different but similar (about what I am doing...)

## Unbalanced Private Set Union

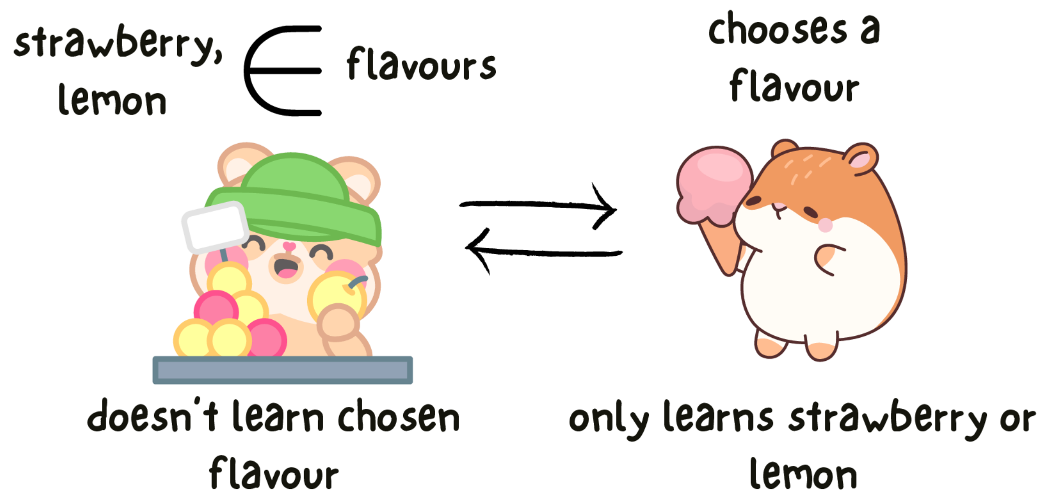


Union is computed without letting the server learn anything about the client, except the full union

## Oblivious Key - Value Store




## oblivious transfers




Oblivious transfer lets one party add their unique elements to a shared union without revealing what they're adding – or learning what the other party holds.

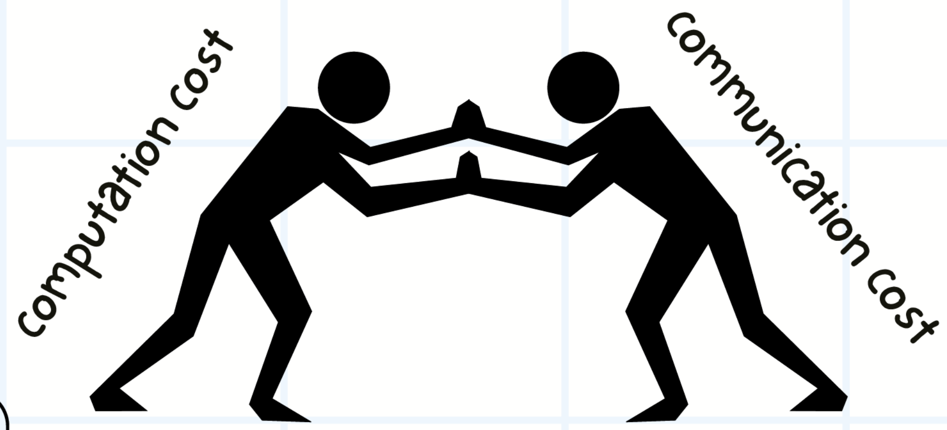
# Conclusion

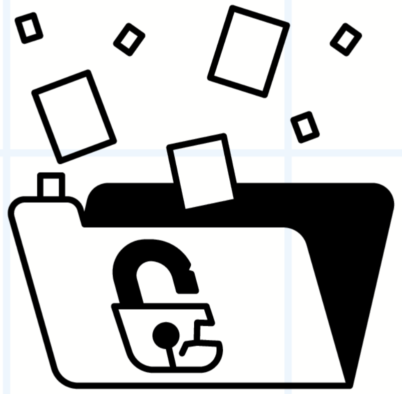
Fuzzy matching is actually a very useful approach for real-world privacy

It is still an early research direction

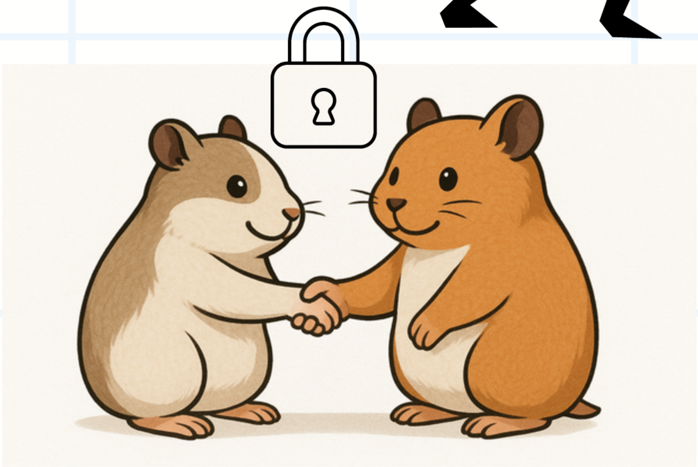
the communication and computational cost is hard to optimize

Many promising protocols, but open challenges remain



leakage models



secure protocols

various attack models

malicious parties

# Conclusion

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Many promising protocols, but open challenges remain